Why This Matters for Evolutionary Computation

Key takeaways and OEE implications from the Genesis research core

1

Fitness-Free Evolution is Viable, but Requires Representational Growth

Selection driven purely by viability regulation and physical constraints keeps populations active.
However, breaking complexity ceilings requires indirect developmental layout (CPPN) + speciation protection.

2

Decoupled Structure and Function Demands Latent Complexity Metrics

Evolutionary runs show a lag between topological expansion (nodes) and functional behavior.
We need quantitative diagnostic suites (like PNCT) to separate active functional complexity from neutral bloat.

3

Standardized OEE Benchmarks: Reusable Sham-Control Protocol

Our sham-control framework isolates ecological feedback loops from direct selection pressure.
This provides a rigorous verification and debugging template for open-ended evolution in any environment.